

Q. Explain the concepts of Microinstruction Sequencing and Execution in a Microprogrammed Control Unit. What is Control Memory?

📌 **Definition to Remember**

In a Microprogrammed Control Unit, a single machine instruction is executed by running a series of smaller steps called **Microinstructions** (collectively forming a **Microprogram**). These microprograms are stored in a dedicated ROM called **Control Memory**.

1. Control Memory

- It is a fast, Read-Only Memory (ROM) located inside the CPU, completely separate from main system memory.
- **Purpose:** Exclusively stores microprograms. When an instruction is decoded, it provides the starting address of that instruction's microprogram in the Control Memory.

2. Microinstruction Execution

A microinstruction is basically a control word. The bits in this word directly translate to physical control lines in the CPU.

* **Execution:** The microinstruction is loaded into a **Microinstruction Buffer**. Its bits directly trigger hardware paths (e.g., a bit **1** might open a register gate, while another bit commands the ALU to Add).

* Multiple control signals are activated simultaneously in one clock cycle just by loading this single control word.

3. Microinstruction Sequencing

Sequencing is the process of determining the address of the *next* microinstruction to fetch from Control Memory. The hardware unit responsible is the **Sequencer**.

The Sequencer determines the next address using:

1. **Next Address Field:** The current microinstruction explicitly contains the address of the next one.
2. **Branching (Conditional):** Based on ALU status flags (Zero/Carry), the sequencer can branch to a different microinstruction, enabling loops and logic.
3. **Opcode Mapping:** The initial machine instruction opcode is mapped to the starting address of the microprogram.
4. **Subroutines:** Microprograms can call subroutines and return using a Micro-Program Counter (Micro-PC).

4. Workflow Loop

1. Map Machine Instruction Opcode → Control Memory starting address.
2. Sequencer fetches Microinstruction.
3. Execution logic generates control signals.
4. Sequencer determines next address.
5. Loop continues until an **End** microinstruction is reached.

★ Must-Write Points (for 10 marks)

1. **Microprogram:** A sequence of microinstructions that execute one machine instruction.
2. **Control Memory:** A dedicated, fast ROM inside the CPU used solely to store microprograms.
3. **Execution:** A microinstruction acts as a control word; its bits directly activate CPU control lines (gates/ALU).
4. **Sequencing:** The process of finding the address of the next microinstruction.
5. The **Sequencer** determines the next address.
6. Sequencing methods: **Next Address Field**, **Conditional Branching** (based on flags), **Opcode Mapping**, and **Subroutines**.
7. The execution is a loop of fetching, signaling hardware, and determining the next micro-address until finished.

⚡ Quick Recall

Microprogram (steps for 1 instruction) → Control Memory (ROM storing steps) → Execution (bits turn into hardware signals) → Sequencing (finding next step via Opcode/Next Field/Branching)